Gaurav Kumar

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Research Experience

Sutton Lab — **Columbia, SC**
Graduate Research Assistant | Python, HPC, Rdkit, ASE — **01/2024 – Present**

- Leveraged generative diffusion models to propose molecular structures, addressing combinatorial scaling challenges and enhancing material structure prediction efficiency.
- Developed tree-based models to accurately predict JV Reverse Scan PCE of Perovskite Materials and fine-tuned encoder based model on molecular structures for precise band gap predictions.

Work Experience

Yojak — **Gurgaon, India**
Software Engineer | Django, React, React Native, Docker, Elasticsearch — **05/2022 – 12/2023**

- Designed and developed a React Native cross-platform e-commerce app with QR SKU scanning and real-time stock updates, boosting user engagement by 35% and reducing order processing time while ensuring transaction fault tolerance.
- Architected and implemented an integrated ERP system in an agile environment, consolidating order processing, accounting, and supply chain management. Automated 15+ manual processes, saving over 40 hours weekly and enhancing operational efficiency.
- Led the creation of automated deployment pipelines using AWS (CloudFront, S3) and Docker for backend and frontend applications. Implemented caching strategies that reduced deployment time by 50%.

Fractal Analytics — **Mumbai, India**
Data Scientist | Python, Jenkins, AWS SageMaker, Git — **10/2020 – 05/2022**

- Built a Customer Lifetime Value (CLV) model for a retail client, optimizing marketing spend across multiple channels and enhancing ROI by 1%, resulting in a $10M increase in annual revenue.
- Developed and deployed an implicit collaborative filtering recommendation model and engineered a seasonal prioritization multiplier, resulting in a 7% ($100k/month) increase in sales validated through A/B testing.
- Increased lead conversion rate by 20% by developing a lead scoring model using tree-based algorithms and deploying it in real-time on Kubernetes with FastAPI.

Technical Skills

Languages: Python, C++, Rust, Go, JavaScript, TypeScript
Frameworks: Django, Django Rest Framework, React, React Native, REST API, Node.js, Express, Next.js
Databases: MySQL, PostgreSQL, MongoDB, Redis, SQLAlchemy
Generative AI: GANs, VAE, Diffusion Models, LLMs, Llama, Anthropic, OpenAI, RAG
Cloud Platforms: GCP, AWS (EC2, S3, SQS, Lambda, Cloudwatch, ECR), Azure
DevOps: Docker, GitHub, GitHub Actions, Kubernetes, Linkerd, Grafana, Kibana, Logstash, Kafka
Developer Tools: Git, Linux, Bash, Postman, Firebase

Education

University of South Carolina — **Columbia, SC**
Masters in Computer Science — GPA: 4.0 / 4.0 — **01/2024 – 05/2025**
Indian Institute of Technology Roorkee — **Roorkee, India**
Bachelors in Engineering Physics — GPA: 7.03 / 10 — **07/2016 – 05/2020**

Projects

Omniport | *React, Django, Django Rest, TypeScript* — **07/2018 – 08/2020**

- Developed the React frontend including core and services of Omniport. It is an end-to-end platform and app ecosystem for institutes to deploy their web portal, by and for the students of IIT Roorkee.

Papers

Do Voters Get the Information They Want? Understanding Authentic Voter FAQs in the US and How to Improve for Informed Electoral Participation arXiv preprint arXiv:2412.15273

Vipula Rawte, Deja Scott, **Gaurav Kumar**, *Aishneet Juneja, Bharat Yaddanapalli, Biplav Srivastava*

Extracurriculars

5th Big Data Health Science Student Case Comptetion — **01/2024**

- Winner of 5th National Big Data Health Science Student Case Competition for developing a model predicting sleep stages accurately.